

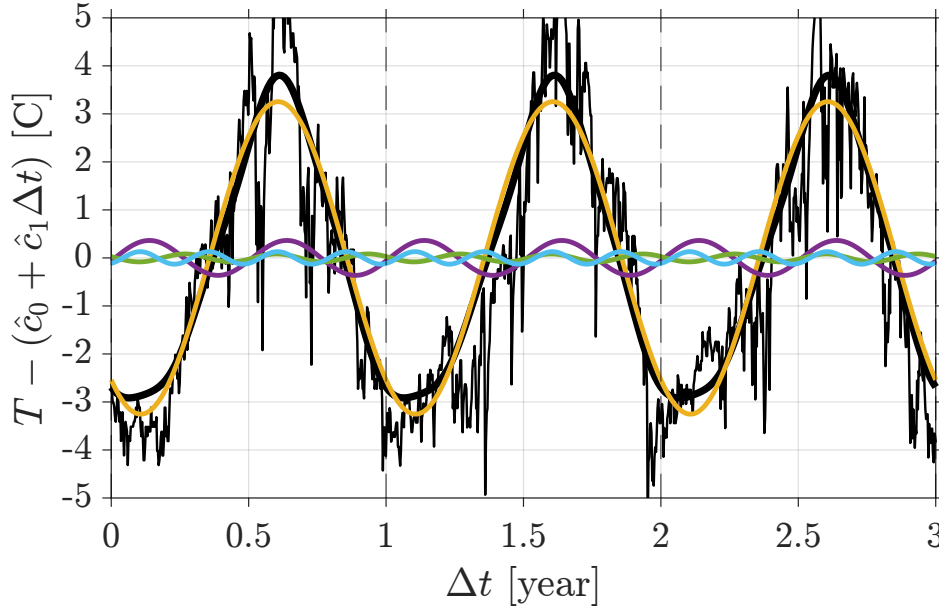


Figure 1: Complex-periodic signal (thick black) composed of sinusoidal functions with fundamental frequency $f_0 = 1$ cycle/year and modes 1-4 (colors) with amplitudes determined via least-squares fit to the SIO-Pier surface temperature record (thin-black).

Reading: B&P {11.2.2,11.5}, E&T {5.6-5.6.3}

In the previous module, we conducted regression analysis on the SIO-pier surface temperature, ending with the least-squares algorithm estimates for coefficients $\{\hat{a}, \hat{b}, \hat{c}\}$ in the model,

$$T = c_0 + c_1 \Delta t + \sum_{k=1}^K [a_k \cos(k \omega_0 \Delta t) + b_k \sin(k \omega_0 \Delta t)], \quad (1)$$

with $\omega_0 = 2\pi/(1 \text{ year})$ and $K = 4$ modes. Our focus was on the nitty-gritty details of the algorithm and uncertainties, not necessarily whether the choice of fundamental period or number of modes was optimal. In particular, we assumed that the periodic signal was either annual or sub-annual with the same fundamental angular frequency ω_0 . Here, we will develop tools to evaluate which time-scales contribute to the variability, and a way of quantifying how much of the variability is associated with the corresponding time-scale.

Before charging ahead, let's evaluate the periodic component of our SIO-pier temperature model (Figure 1). Plotting each mode k of the sum in (1) illustrates that the majority of the amplitude of the model fit is from just the first two modes 1 and 2 cycles per year (yellow and purple, respectively). To do this more directly, we can simplify the summed $\cos()$ + $\sin()$ terms by first substituting,

$$\left. \begin{aligned} a_k &= A_k \sin(\phi_k) \\ b_k &= A_k \cos(\phi_k) \end{aligned} \right\} \quad \text{with} \quad \left\{ \begin{aligned} A_k &= \sqrt{a_k^2 + b_k^2} \\ \phi_k &= \tan^{-1}(b_k/a_k) \end{aligned} \right.$$

and applying the trig-identity, $\cos(\omega_0 \Delta t - \phi_k) = \cos(\phi_k) \cos(\omega_0 \Delta t) + \sin(\phi_k) \sin(\omega_0 \Delta t)$, which gives,

$$a_k \cos(k \omega_0 \Delta t) + b_k \sin(k \omega_0 \Delta t) = A_k \cos(k \omega_0 \Delta t - \phi_k). \quad (2)$$

So, each of our modes can be described by either $\{a_k, b_k\}$ or a single amplitude A_k and initial phase ϕ_k . By plotting the amplitude and phase versus mode number (Figure 2), it becomes clear just how little temperature variability is described by the inclusion of modes 3 and 4 cycles per year. Here, we will develop formal methods to analyze variability versus time-scale.

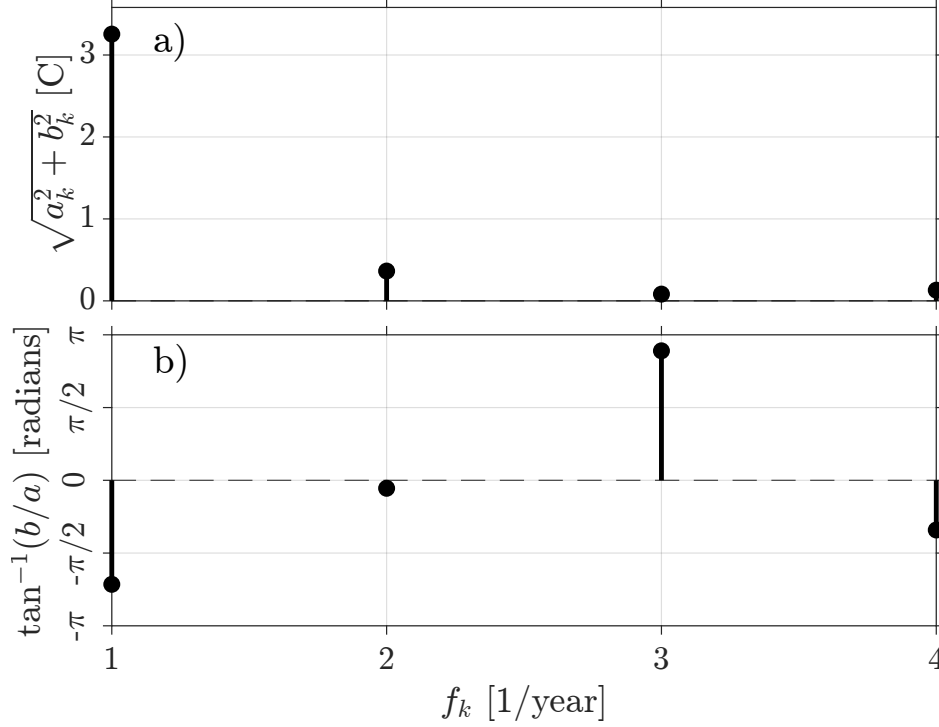


Figure 2: a) Sinusoidal amplitudes and b) phases from least-squares fit to the SIO-pier temperature record.

1 Fourier Series

We have previously seen that any complex-periodic signal $x(t)$ can be represented by the Fourier series,

$$x(t) = \frac{a_0}{2} + \sum_{k=1}^{\infty} [a_k \cos(k \omega_0 t) + b_k \sin(k \omega_0 t)] \quad (3)$$

with fundamental period $\mathcal{T}_0$, for which $\omega_0 = 2\pi/\mathcal{T}_0$. Recall, the fundamental period satisfies $x(t) = x(t + n \mathcal{T}_0)$ for all integer n (see Lecture 2). Our previous least-squares model was a truncated Fourier series with a linear trend term. In the continuous limit, the coefficients $\{a_k, b_k\}$ are determined by the inner-product between the signal and corresponding mode, *i.e.*,

$$a_k = \frac{2}{\mathcal{T}_0} \int_0^{\mathcal{T}_0} x(t) \cos(k \omega_0 t) dt, \quad (4)$$

$$b_k = \frac{2}{\mathcal{T}_0} \int_0^{\mathcal{T}_0} x(t) \sin(k \omega_0 t) dt. \quad (5)$$

Are you wondering why we didn't use a discrete version of the above relations to estimate the modal coefficients? The reason is that the above relations are only true if the record $\mathcal{T}$ is an integer multiple of $\mathcal{T}_0$ (or is very large $\mathcal{T} \gg \mathcal{T}_0$), in which case,

$$\int_0^{\mathcal{T}} \cos(k \omega_0 t) \cos(l \omega_0 t) dt = 0 \quad \text{if } l \neq k, \quad (6)$$

and similarly for $\sin(k \dots) \sin(l \dots)$. The cross terms $\sin(k \dots) \cos(l \dots)$ are always zero when integrated over a fundamental period $\mathcal{T}_0$. If the record is shorter than the fundamental period $\mathcal{T} < \mathcal{T}_0$, the modes are not independent. In these situations, the least-squares algorithm is ideal because it accounts for potential covariance between the component functions. We can evaluate the skill of the Fourier coefficient estimates as a function of the record length by trimming our periodic solution shown in Figure 1 and Figure 2. Coefficients

$\{a, b\}$ were estimated using the Fourier method (red, Figure 3) and least-squares method (black, Figure 3) on records of varying length from $\mathcal{T} = 0.1$ year to 2 year and the corresponding skill-score recorded. The Fourier method is trash until the record length becomes comparable to $\mathcal{T}_0 = 1$ year. It decreases slightly when $\mathcal{T}$ increases beyond 1 year but is as good as least-squares beyond $\mathcal{T} = 1.5$ years. In contrast, the least-squares method is nearly perfect over the whole range. This is why tidal harmonic analysis generally uses a least-squares method to estimate constituent amplitude and phase.

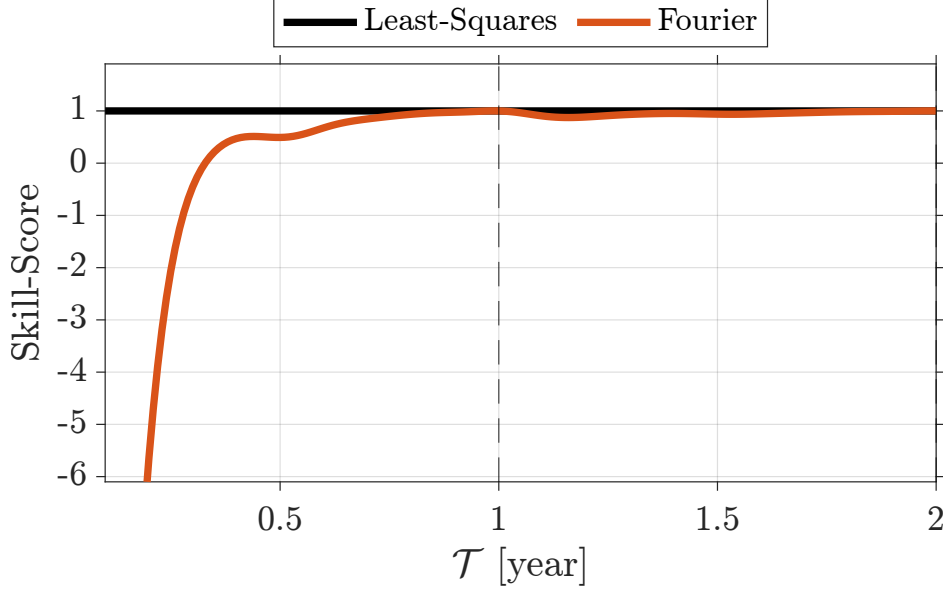


Figure 3: Comparison of skill-scores for least-squares fit (black) and Fourier coefficients (red) versus record length $\mathcal{T}$. The periods of the sinusoids are integer multiples of 1 year.

2 Fourier Transform

To conduct an objective analysis of the scales (*e.g.*, time $\mathcal{T}$ or length λ) contributing to a record, we must first transform our discrete record from physical space to frequency ω and wavenumber k space,

$$\begin{aligned} q(t) &\rightarrow Q(\omega) \quad \text{with } \omega = \frac{2\pi}{\mathcal{T}}, \\ r(l) &\rightarrow R(k) \quad \text{with } k = \frac{2\pi}{\lambda}. \end{aligned}$$

To do so, we'll use the Fourier transform (FT), defined for continuous variables as,

$$X(\omega) = \int_{-\infty}^{\infty} x(t) \exp(-i\omega t) dt \quad (7)$$

where $i = \sqrt{-1}$ is the imaginary unit, and the complex exponential is,

$$\exp(\pm i\theta) = \cos(\theta) \pm i \sin(\theta).$$

The FT variable has both real and imaginary components, $X(\omega) = X_{\Re}(\omega) + iX_{\Im}(\omega)$, that may be represented by an amplitude and phase similar to the Fourier series coefficients,

$$\begin{aligned} A^2 &= X^* X = (X_{\Re} - iX_{\Im})(X_{\Re} + iX_{\Im}) = X_{\Re}^2 + X_{\Im}^2, \\ \phi &= \tan^{-1}(X_{\Im}/X_{\Re}), \end{aligned}$$

where the asterisk $()^*$ indicates complex conjugations $i \rightarrow -i$. Both are functions of ω , with the squared-amplitude A^2 describing how much energy (or variance) is associated with ω and the phase ϕ the initial

phase at $t = 0$. The FT is complemented by the inverse-Fourier transform (IFT), which is defined as,

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) \exp(i\omega t) d\omega = \int_{-\infty}^{\infty} X(2\pi f) \exp(i2\pi ft) df. \quad (8)$$

If the signal $x(t)$ is real $x(t) = x^*(t)$, then the FT has the property $X(\omega) = X^*(-\omega)$ and we only need to retain the positive portion of the transformation. The proof of this begins with the complex-conjugate of the IFT,

$$\begin{aligned} x^*(t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} [X(\omega) \exp(i\omega t)]^* d\omega \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X^*(\omega) \exp(-i\omega t) d\omega, \quad \text{substitute } \omega \rightarrow -\omega, \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X^*(-\omega) \exp(i\omega t) d\omega \end{aligned}$$

which requires $X^*(-\omega) = X(\omega)$ for $x^*(t) = x(t)$.

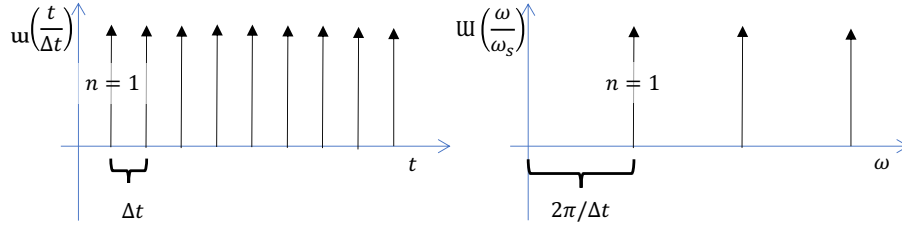


Figure 4: Example of $\mathfrak{w}(t/\Delta t)$ and $\mathfrak{W}(\omega/\omega_s)$ where $\omega_s = 2\pi/\Delta t$ and assuming $\Delta t \ll 1$.

2.1 Sampling Function

As usual, the situation is complicated by having both a discrete and finite length record. In order to understand these effects mathematically, we introduce the Shah-Functions (or sampling functions),

$$\mathfrak{w}\left(\frac{t}{\Delta t}\right) = \Delta t \sum_{n=-\infty}^{\infty} \delta(t - n\Delta t), \quad (9)$$

$$\mathfrak{W}\left(\frac{\omega}{\omega_s}\right) = \omega_s \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_s), \quad \text{with } \omega_s = \frac{2\pi}{\Delta t}, \quad (10)$$

represented in physical and frequency space, respectively (Figure 4). The component function is the delta-function $\delta(x - a)$ (black arrows, Figure 4), which is zero everywhere except at $x = a$ and has the relevant properties,

$$\begin{aligned} \int_{-\infty}^{\infty} \delta(x) dx &= 1, \\ \int_{-\infty}^{\infty} f(x) \delta(x - a) dx &= f(a), \end{aligned}$$

among others. The second relationship is rationalized by allowing $\delta(x - a)$ to be non-zero over the infinitesimal interval $a - \epsilon \leq x \leq a + \epsilon$ where the integral is approximately,

$$\int_{a-\epsilon}^{a+\epsilon} f(x) \delta(x - a) dx \approx f(a) \int_{a-\epsilon}^{a+\epsilon} \delta(x - a) dx = f(a).$$

The above approximation becomes exact in the limit $\epsilon \rightarrow 0$. If the sample interval Δt is small, the delta-functions are close together in physical space and widely-separated in frequency space, and vice-versa.

To extend our FT analysis to the discrete case, we first scale the sampling function with our continuous signal,

$$x_s(t) = x(t) \text{m}(t/\Delta t).$$

Applying the FT to this expression gives the discrete Fourier transform,

$$\begin{aligned} X_s(\omega) &= \int_{-\infty}^{\infty} x_s(t) \exp(-i\omega t) dt \\ &= \int_{-\infty}^{\infty} \sum_{n=-\infty}^{\infty} \delta(t - n\Delta t) x(t) \exp(-i\omega t) \Delta t dt \\ &= \sum_{n=-\infty}^{\infty} x(n\Delta t) \exp(-i\omega n\Delta t) \Delta t, \end{aligned} \quad (11)$$

which converges on $X(\omega)$ in the limit $\Delta t \rightarrow 0$. However, we are limited to N realizations of $x(t)$, resulting in the truncated series,

$$X_N(\omega_j) = \sum_{n=0}^{N-1} x(n\Delta t) \exp(-i\omega_j n\Delta t) \Delta t, \quad \text{with } \omega_j = j \frac{2\pi}{N\Delta t}. \quad (12)$$

The angular frequencies are quantized because we can only decompose our signal into modes that fit our range of observations with an integer number of oscillations. Recall, the Fourier series coefficients were optimally fit only when the record length was an integer multiple of the fundamental period (Figure 3). In effect, we impose a periodization condition $x(t + m(N\Delta t)) = x(t)$ for integer m , which is not likely to be true. We'll discuss this later.

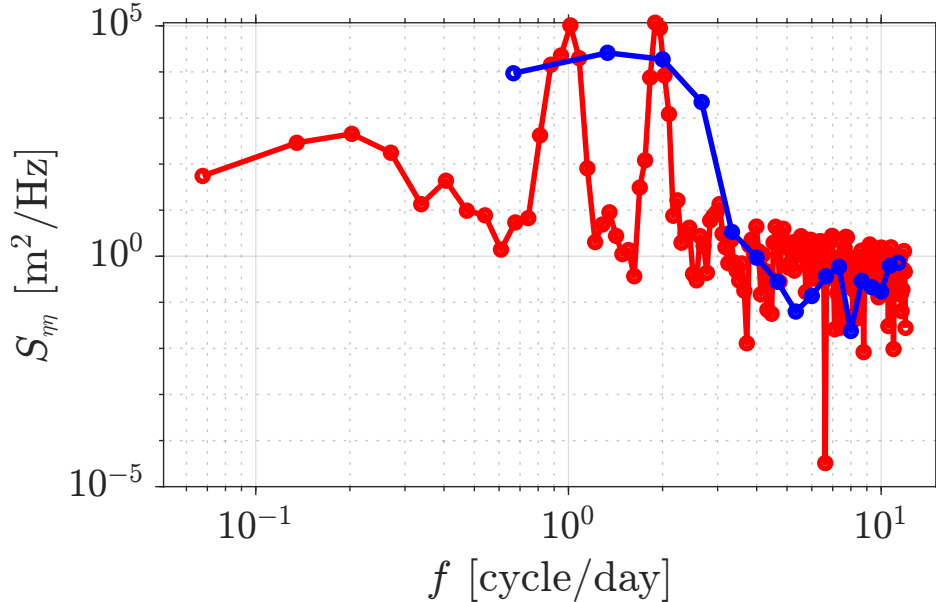


Figure 5: Squared amplitude of Fourier transformed SIO-pier tidal records using $\mathcal{T}_0 = 1.5$ days (blue) and $\mathcal{T}_0 = 14.78$ days (red). The coarse spectral resolution (blue) of $\Delta f = 2/3$ cycles/day does not resolve the diurnal and semi-diurnal tidal constituents, 1 and 2 cycles/day, respectively.

2.2 Fundamental Period & Rayleigh Criterion (light)

The fundamental period $\mathcal{T}_0 = N\Delta t$ and frequency $f_0 = 1/(N\Delta t)$ are the longest time-scales and lowest frequencies that we can resolve, respectively. The fundamental period is also our spectral resolution, *i.e.*,

each mode is separated by $\Delta f = f_{j+1} - f_j = f_0$. As a consequence, if we hope to distinguish between two different signals that are delta-functions in spectral space, *i.e.*,

$$X(\omega) = X_1(\omega) + X_2(\omega), \quad \text{with,} \quad \begin{cases} X_1 &= A_1 \delta(\omega - 2\pi f_1), \\ X_2 &= A_2 \delta(\omega - 2\pi f_2), \end{cases} \quad (13)$$

then we must have at least one sample between them where $|X| < \min(A_1, A_2)$, otherwise they appear as one single hump (*e.g.*, blue, Figure 5). This condition is minimally satisfied if,

$$f_0 < |f_2 - f_1|, \quad (14)$$

which puts a requirement on the record length necessary to resolve these signals as,

$$N > f_s \frac{T_1 T_2}{|T_1 - T_2|} \quad (15)$$

for a given sample rate $f_s = 1/\Delta t$. The above relations are known as Rayleigh criteria, and rely on the input signals being delta functions and the first minima of our windowing function (we'll see this in detail later) occurring at $\pm \Delta f$. If the spectral peaks are themselves wider and do not satisfy the Rayleigh criteria, no amount of sampling will distinguish them.

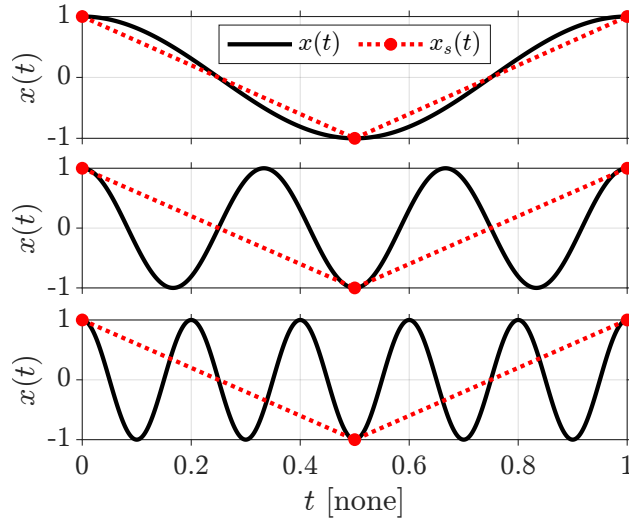


Figure 6: Three example signals $x(t)$ (black) that project onto the sampled record x_s (red) at the Nyquist frequency f_{Nyq} . Although the top is perfectly resolved, a simple phase shift (delaying sampling by ± 0.25) would result in zero amplitude. Thus, all are poorly sampled.

2.3 Nyquist Frequency & Aliasing

The angular frequencies ω_j in (12) are also limited in magnitude to $j \leq N/2$,

$$\max[\omega_j] = j_{\max} \frac{2\pi}{N\Delta t} = \frac{2\pi}{2\Delta t}, \quad (16)$$

which means the shortest resolvable time-scale is $\mathcal{T}_{\min} = 2\Delta t$ (top, Figure 6). This is called the Nyquist frequency limit,

$$f_{\text{Nyq}} = \frac{1}{2\Delta t}. \quad (17)$$

Signals at frequencies higher than f_N are under-sampled and cannot be accurately resolved. If we consider the complex exponential in (12) at values $j = 1$ and $j = N/2 + 1$,

$$\begin{aligned} j = 1 : & \quad \exp(-i\omega_0 n \Delta t), \\ j = \frac{N}{2} + 1 : & \quad \exp(-i\omega_0 (\frac{N}{2} + 1) n \Delta t), \\ & \quad \exp(-i\omega_0 n \Delta t) \exp(-if_{\text{Nyq}} n \Delta t), \end{aligned}$$

the $j = N/2 + 1$ exponential oscillates at the Nyquist frequency f_{Nyq} with an amplitude that is modulated at the fundamental frequency f_0 .

A problem with under-resolving signals above f_{Nyq} is that their energy (variance) does not disappear, it is folded (or aliased) into lower frequency modes. There are two ways of rationalizing how this happens, the second will be discussed in the next section. First, consider the requirement of a complex periodic signal $x(t + m\mathcal{T}_0) = x(t)$. In the case of a discrete record, we've sampled a signal at a rate of $f_s = 1/\Delta t$ for a period of $\mathcal{T}_0 = N\Delta t$ and our assumption is that the record exactly repeats itself every $\mathcal{T}_0$, and for any given mode k ,

$$\begin{aligned}\exp(i k 2\pi f_0 n \Delta t) &= \exp[i k 2\pi f_0 (n \Delta t + m N \Delta t)], \\ &= \exp[i k 2\pi (f_0 n \Delta t + m)].\end{aligned}$$

If instead, we reverse the above statement by imagining a rapidly oscillating signal f that repeats itself many times between samples $f > f_s$, then we might sample successive cycles at just the right interval to get an apparent low-frequency oscillation,

$$\begin{aligned}\exp(i 2\pi f n \Delta t) &= \exp[i 2\pi (f_A n \Delta t + m * n)], \\ &= \exp[i 2\pi \underbrace{(f_A + m f_s)}_f n \Delta t],\end{aligned}$$

where f_A is the aliased frequency of a signal at frequency $f > f_{\text{Nyq}}$. If f_A is one of our resolved modes, then any variability at f will be incorrectly attributed to f_A . If f is known, the aliasing frequencies are those that satisfy,

$$f_A = |f - m f_s|, \quad \text{for } m = \pm 1, \pm 2, \dots \quad (18)$$

In Figure 6, three signals $f = \{1, 3, 5\}$ (black) are shown that all map to $f_A = 1$ for a sample rate of $f_s = 2$, *e.g.*,

$$\begin{aligned}m = 1 : \quad & f_A = |1 - 2| = 1 \\ m = 2 : \quad & f_A = |3 - 4| = 1 \\ m = 3 : \quad & f_A = |5 - 6| = 1\end{aligned}$$

2.3.1 Fourier Transform Product/Convolution Theorem

One very useful property of the FT is that when you transform the product of two functions, say $\text{FT}[x(t) \text{III}(t/\Delta t)]$, the result is equivalent to the convolution of their individual transformations, that is,

$$\begin{aligned}\text{FT}[x(t) \text{III}(t')] &= \int_{t'} \underbrace{x(t')}_{\downarrow} \text{III}(t') \exp(-i\omega t') dt' \\ &= \int_{t'} \int_{\omega'} \overbrace{X(\omega') \exp(i\omega' t')}^{\downarrow} \text{III}(t') \exp(-i\omega t') dt' \\ &= \int_{\omega'} X(\omega') \underbrace{\int_{t'} \text{III}(t') \exp(-i(\omega - \omega') t') dt'}_{\downarrow} d\omega' \\ &= \int_{\omega'} X(\omega') \overbrace{\text{III}(\omega - \omega')}^{\downarrow} d\omega' \\ &= X \star \text{III},\end{aligned} \quad (19)$$

where the integral ranges are from $-\infty$ to ∞ . This property is useful for filtering time-series. In our analysis of joint PDFs we saw that convolving two jagged distributions resulted in a smoother function. We will later create filter functions that target certain frequency ranges, and that can be applied to data by either

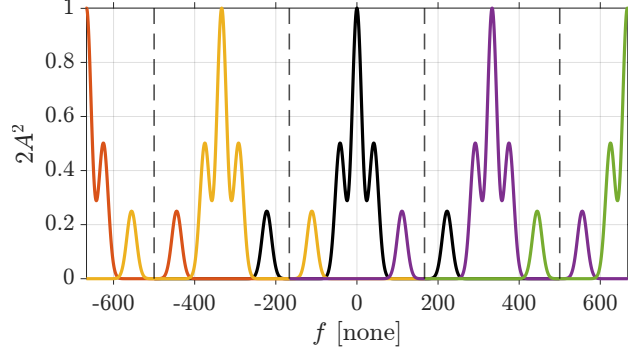


Figure 7: Example of a true spectrum (black) and the aliased spectrum (colors) due to convolution of the sampling function with the true spectrum.

convolving in the time-domain or transforming to the frequency domain, direct multiplication, then inverse transforming them.

Aliasing can also be viewed as a consequence of the convolution property of the FT. Our discrete Fourier transform is equivalent to the convolution of the true transform and the FT of our sampling function,

$$\begin{aligned}
X_s(\omega) &= X \star \text{III}, \\
&= \int_{\omega'} X(\omega') \text{III}(\omega - \omega') d\omega', \\
&= \int_{\omega'} X(\omega') \sum_n \delta(\omega - \omega' - n\omega_s) d\omega', \\
&= \int_{\omega'} X(\omega - \omega') \sum_n \delta(\omega' - n\omega_s) d\omega', \\
&= \sum_n X(\omega - n\omega_s).
\end{aligned} \tag{20}$$

In words, the discrete spectrum is the sum of infinitely many copies of the true spectrum, each shifted by $\omega_s = 2\pi/\Delta t$. So, if the true signal has energy at frequencies higher than $f_{\text{Nyq}} = 1/(2\Delta t)$ then these signals from neighboring copies will be aliased into our resolvable region. To illustrate this, consider a true spectrum composed of Gaussian bumps located at $f = \{0, 42, 222\}$ (black, Figure 7) and the neighboring copies from a discrete spectrum having $f_{\text{Nyq}} = 3/4\max(f) = 166$ (colors). In the resolved range of $\pm f_{\text{Nyq}}$ (vertical black dashed lines), the high frequency peak of the nearest neighboring functions will be aliased, in this case creating a spurious peak at,

$$f_A = \left| \frac{4}{3}f_{\text{Nyq}} - 2f_{\text{Nyq}} \right| = \frac{2}{3}f_{\text{Nyq}} = 111.$$

3 Spectral Amplitude & Parseval's Theorem

We have seen that the magnitude of the FT coefficients $A_j^2 = X_j^* X_j$, where $X_j = X_N(\omega_j)$ following (12), give an indication of how much observed variability is described by the j^{th} mode. To quantify this contribution, we'll have to normalize our transformation and inverse transformations so that they recover the same variance (energy). This property of the Fourier transform is called Parseval's theorem: for a real function $x(t) = x^*(t)$, the integrated variance in physical space is equivalent to that in frequency/wavenumber space,

$$\int_{t'} x(t) x(t) dt = \int_{f'} X^*(f') X(f') df', \tag{21}$$

where we've switched to integrating over frequency space $f = \omega/(2\pi)$ for future simplicity. Let's prove the above relation for a discrete variable $x_n = x(t_n)$, with $t_n = n\Delta t$, beginning with the left-hand-side of (21),

$$\begin{aligned} \int_{t'} x(t) x(t) dt &\rightarrow \sum_n x_n x_n \Delta t, \\ &= (x_1^2 + x_2^2 + \dots x_N^2) \Delta t, \\ &= (N\Delta t) \hat{\sigma}_x^2, \end{aligned} \tag{22}$$

where $\hat{\sigma}_x^2$ is the biased sample variance estimator and I've assumed the record has zero mean $\bar{x} = 0$. Following a similar procedure for the discrete form for the right-hand-side of (21), we can use a little trick substitution to prove equivalence,

$$\begin{aligned} \int_{f'} X^*(f') X(f') df' &\rightarrow \sum_j X_j^* \underbrace{X_j}_{\searrow} \Delta f, \\ &= \sum_j X_j^* \overbrace{\sum_n x_n \exp(-i2\pi f_j t_n) \Delta t}^{\Delta f}, \\ &= \sum_n x_n \underbrace{\sum_j X_j^* \exp(-i2\pi f_j t_n) \Delta f}_{\swarrow} \Delta t, \\ &= \sum_n x_n \overbrace{x_n^*}^{\swarrow} \Delta t = (N\Delta t) \hat{\sigma}_x^2. \end{aligned} \tag{23}$$

If we want our amplitudes A_j to quantify the amount of variance within each mode, we must properly re-normalize them. Most computer algorithms that calculate DFT and their inverse do not account for the factors of Δt . So, X_j computed using standard algorithms must be multiplied by Δt for the above relations to hold. The two most common normalizations are,

$$\begin{aligned} i) \quad A_j^2 &= \frac{X_j^* X_j}{N^2}, & \text{with units } [x^2], \text{ and} \\ ii) \quad A_j^2 &= \frac{X_j^* X_j \Delta t}{N}, & \text{with units } \left[\frac{x^2}{f_0} \right]. \end{aligned}$$

If our observations are real, then we can double the amplitude $A \rightarrow 2A$ and retain only positive frequencies. Choice (ii) is ideal when comparing spectral plots from records with different spectral resolution (*e.g.*, Figure 5). If instead, we used option (i) (*i.e.*, normalizing by N^2) the 1.5 – day amplitudes would be significantly higher ($\approx 9\times$) than the 14.78 – day record, so visually it would appear there's extra variance in the short record. Henceforth, we will use normalization (ii) and assume the data is real, for which the observed variance is,

$$\hat{\sigma}_x^2 = \sum_{j=0}^{N/2} 2A_j^2 f_0, \tag{24}$$

where $\Delta f = f_0 = 1/(N\Delta t)$. You'll notice that we previously defined the indices to run from $n = 0$ to $N - 1$, here we only go up to the Nyquist frequency. One final note, the zero-frequency contains the variance in the sample mean, because $\exp(i0) = 1$, and this is also the first term $a_0/2 = \bar{x}$ in the Fourier series representation (3). For reasons we'll address in the coming sections, we'll be removing the mean and linear trend prior to transforming our data.

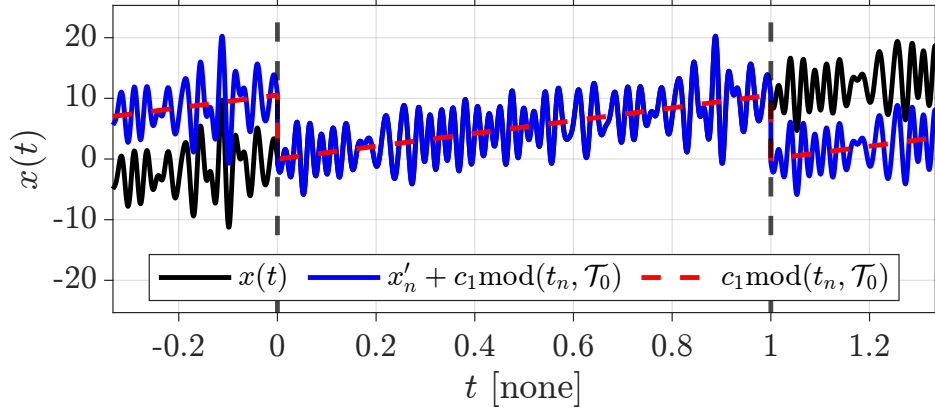


Figure 8: Example of a complex-periodic signal with a linear trend (solid black) sampled between $0 \leq t \leq 1$ (vertical dashed black) and the periodized version of the sample record (blue) showing the imposed saw-tooth characteristic (dashed red).

4 Periodization

4.1 Detrending

As we've already discovered, the N -point discrete Fourier transform has inherent limitations in resolution (f_0) and aliasing ($f > f_{\text{Nyq}} \rightarrow f_A$). Here, we're going to address another limitation associated with the periodization of the signal $x(t + mT_0) = x(t)$, with $T_0 = N\Delta t$, when there is a trend in the discrete record. If we enforce the periodization requirement, then we are suggesting the record $x(t)$ has a saw tooth shape (Figure 8), formally $x(t_n) = x'_n + c_1 \text{mod}(t, T_0)$ where x' is a zero-mean oscillating component and the modulus function is defined as $\text{mod}(a, b) = a - mb$ with $m = \text{floor}(a/b)$ the integer cycle number. If there is truly a long-term trend, this is a terrible representation.

The problem becomes clear when considering the Fourier transform of the underlying linear trend. The periodized trend looks like a saw-tooth, sloping upward to the right (red, Figure 8). To simplify the analysis, we'll change variables,

$$\begin{aligned} x &= \frac{A}{T}t & 0 \leq t \leq T, \\ y &= \frac{A}{T}\gamma & -\frac{T}{2} \leq \gamma \leq \frac{T}{2}, \end{aligned}$$

for which the Fourier series representation is,

$$y = \sum_n c_n \exp\left(in \frac{2\pi}{T} \gamma\right). \quad (25)$$

with Fourier coefficients,

$$\begin{aligned} c_n &= \frac{1}{T} \int_{-T/2}^{T/2} \frac{A}{T} \exp\left(-in \frac{2\pi}{T} \gamma\right) dt, \\ &= \frac{iA}{n2\pi T} \left[\gamma \exp\left(-in \frac{2\pi}{T} \gamma\right) \right]_{-T/2}^{T/2} - \underbrace{\frac{iA}{n2\pi T} \int_{-T/2}^{T/2} \frac{A}{T} \exp\left(-in \frac{2\pi}{T} \gamma\right) dt}_{=0}, \\ &= \frac{iA}{2\pi n} \cos(n\pi) = \frac{iA}{T} \frac{(-1)^n}{n\omega_0}. \end{aligned} \quad (26)$$

The modes $n\omega_0$ occur at integer multiples of the inverse record length $1/T$, and each mode has a decreasing amplitude. The properly scaled amplitude corresponding to x is shown as red arrows in Figure 9. As you

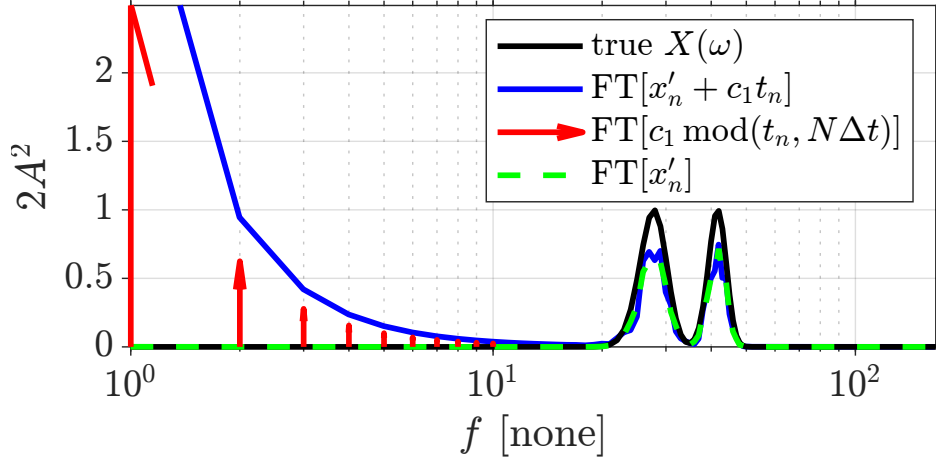


Figure 9: Example spectra computed from original periodic record $x(t)$ (black), a sample record with a linear trend $x(t_n) = x'_n + c_1 t_n$ (blue), a sawtooth function $c_1 \text{mod}(t_n, N\Delta t)$ representing the linear trend (red), and the detrended record x'_n .

can see, the amplitudes of the sample spectra are offset above zero at low frequencies due to the effect of a linear trend. Removing the trend before computing the FT removes this low-frequency bias. A spectrum that slopes upward to the left (low-frequencies) is often called “red” in analogy to red light having long wavelengths and low frequencies. A blue spectrum is tilted upward toward positive frequencies. And white spectra are flat.

4.2 Tapering

A final issue with the imposed periodization of the FT is that the oscillating portion of our record does not likely start and end with the same value, slope, etc. That is, there is likely to be a discontinuity between the end of one segment $x(N\Delta t)$ and the next $x(N\Delta t + 1) = x(1)$, because it is unlikely that every signal in our record has a period $\mathcal{T}$ that satisfies $m\mathcal{T} = N\Delta t$ for integer m . For example, the discrete signal,

$$x(n) = \cos(2\pi/50 n) + 0.5 \cos(2\pi/25 n\Delta t)$$

was used to create two sample sets, x_p having $N_p = 200$ data-points (an integer multiple of fundamental period $\mathcal{T}_0 = 50$) and x_a having $N_a = 188$ data-points (Figure 10). Applying the FT to aperiodic records with such discontinuities leads to “spectral leakage”, where variance is spread from a narrow signal to neighboring

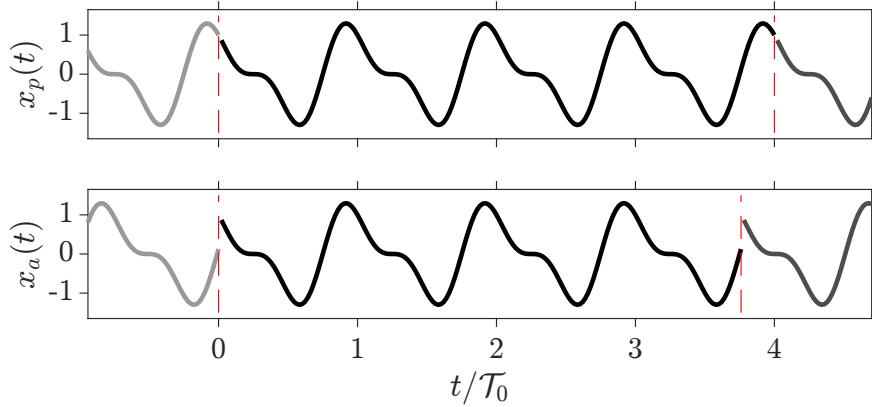


Figure 10: Examples of a periodic record (top) where the sample duration exactly matches an integer multiple of the fundamental period and an aperiodic record (bottom) where it does not. Notice the discontinuity between neighboring copies for the aperiodic record.

frequencies. To account for this, we “taper” our data window to smoothly bring the data to zero at the start and end of the record.

To see this in action, we reconsider our data record to be the product $x_s(t) = x(t)w(t)$ of the true signal $x(t)$ and our record window $w(t)$, where,

$$w(t) = \begin{cases} 0 & \text{for } t < 0 \text{ \& } \mathcal{T}_0 < t, \\ 1 & \text{for } 0 \leq t \leq \mathcal{T}_0 \end{cases} \quad (27)$$

This form of w is called the box-car window (red, Figure 11a). As we’ve seen previously, the FT of a product is equivalent to the convolution of the individual FTs, that is,

$$\text{FT}[x(t)w(t)] = X(\omega) \star W(\omega).$$

To determine the Fourier transform of the box-car window, we’ll make a change of variables $\gamma = t - \mathcal{T}/2$, such that,

$$\begin{aligned} W(\omega) &= \int_{-\infty}^{\infty} w(\gamma) \exp(-i\omega\gamma) d\gamma, \\ &= \int_{-\mathcal{T}/2}^{\mathcal{T}/2} 1 \exp(-i\omega\gamma) d\gamma, \\ &= \left[-\frac{\exp(-i\omega\gamma)}{i\omega} \right]_{-\mathcal{T}/2}^{\mathcal{T}/2}, \\ &= \frac{\exp(i\omega\mathcal{T}/2) - \exp(-i\omega\mathcal{T}/2)}{i\omega}, \end{aligned}$$

which may be simplified using the identity $\sin(\theta) = (\exp(i\theta) - \exp(-i\theta))/(2i)$ and $\omega = 2\pi f$,

$$W(\omega) = \mathcal{T} \frac{\sin(\pi f \mathcal{T})}{\pi f \mathcal{T}} = \mathcal{T} \text{sinc}(f\mathcal{T}). \quad (28)$$

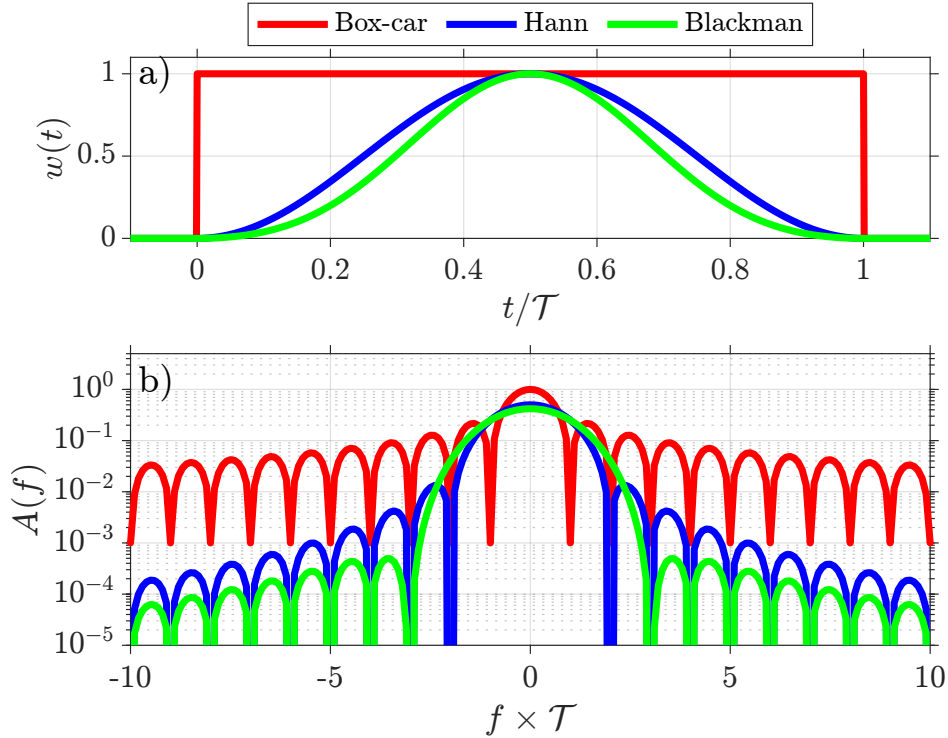


Figure 11: Examples of the standard box-car window (red) a $\cos^2()$ Hann function window (blue) and Blackman window (green) with respect to a) amplitude in the time domain and b) spectral amplitude in the frequency domain.

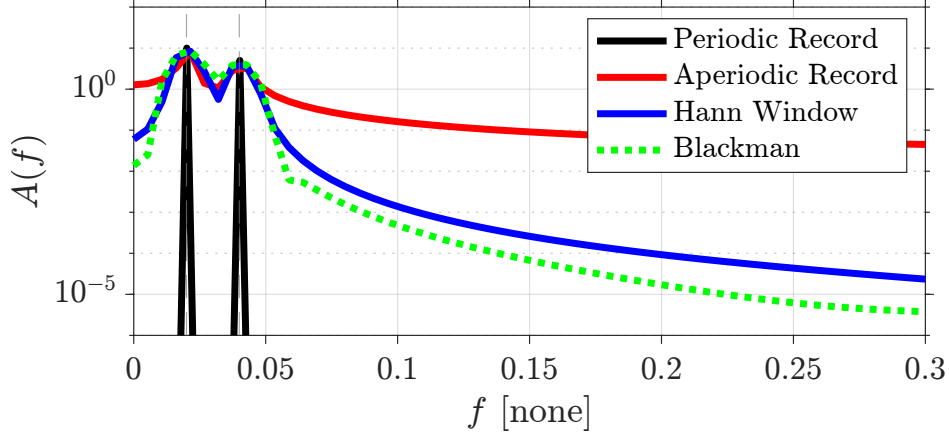


Figure 12: Examples of spectra from a periodic record (black) and aperiodic record (red) using the standard box-car window. Also shown are tapered versions of the aperiodic record using a $\cos^2()$ Hann function window (blue) and a Blackman window (green).

The sinc-function is symmetric, oscillates with zeros at $|fT| = m$ for integer m (not shown). The amplitude of these oscillations decay from $\text{sinc}(0) = 1$ with increasing m (red, Figure 11b).

When the transformed window $W(\omega)$ is convolved with $X(\omega)$, energy spreads out in frequency space due to the side-lobes. The spectral amplitudes of the box-car window for both are significantly different (black and red, respectively, Figure 12), with the periodic amplitudes being spikes at $f = 0.2$ and 0.4 and the aperiodic record having additional energy over the full frequency range. To partially correct for this issue, we premultiply our data-record by a function that tapers smoothly to zero the start and end of the records. The most basic shape is that from half of a $\cos^2()$ cycle, called the Hann function,

$$w(n) = 1 - \cos^2\left(\frac{2\pi}{2N}n\right) = \frac{1}{2} - \frac{1}{2}\cos\left(\frac{2\pi}{N}n\right), \quad \text{for } 1 \leq n \leq N. \quad (29)$$

The above right hand side is a specific form of the generalized cosine tapers,

$$w(n) = a_0 - a_1 \cos\left(\frac{2\pi}{N}n\right) + a_2 \cos\left(\frac{4\pi}{N}n\right) \dots$$

The three term version is called the Blackman function and has coefficients,

$$a_0 = \frac{1 - \alpha}{2} \quad a_1 = \frac{1}{2} \quad a_2 = \frac{\alpha}{2}, \text{ and } \alpha \approx 0.16.$$

The Hann function and Blackman functions have wider main-lobes and small side-lobes. So if you don't mind widening the primary peak, the added reduction in overall spectral leakage is significant (blue and red, Figure 12).

One draw-back about tapering your input record is that it reduces the variance passed through the FT and the amplitudes will not satisfy Parseval's theorem. Addressing this is trivial,

- Record your aperiodic sample variance: s_a^2 ,
- apply your taper: $x_p(n) = w(n)x_a(n)$,
- estimate the variance of the tapered record: s_p^2 ,
- re-scale the tapered record: $x_p = x_p \times (s_a^2/s_p^2)$.

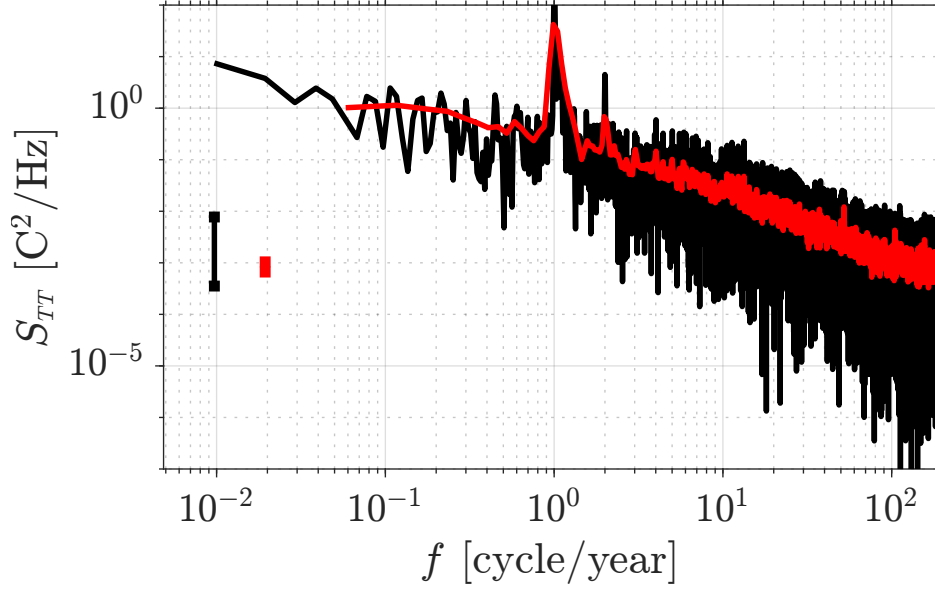


Figure 13: Spectral density of SIO-pier temperature with two degrees of freedom (black) and using $M = 16$ sub-windows with $\beta = 2/3$ overlap and ≈ 64 degrees of freedom. The corresponding confidence intervals are shown as vertical lines on the left side of the axes.

4.3 Confidence Intervals

The confidence interval (CI) for spectral amplitudes are similar to that of variance estimates $\hat{\sigma}^2$. The confidence interval probability statement is,

$$\text{Prob}[|\hat{\phi} - \phi| \leq \delta_\alpha] = 1 - \alpha,$$

which for spectral intensity $\phi = S(f)$ amounts to a frequency dependent interval $\hat{S}(f) - S(f)$. This formulation is rather cumbersome, particularly when plotting because each frequency has its own error magnitude. If instead, we intend on plotting the spectra on a log-scale, then we can log-transform the statistic $\phi \rightarrow \log(S(f))$ which converts the difference to a ratio,

$$(\hat{\phi} - \phi) \rightarrow (\log(\hat{S}) - \log(S)) = \log\left(\frac{\hat{S}(f)}{S(f)}\right).$$

The spectral density $\hat{S}(f_k) = A_k^2$ is a χ^2 variable because it involves a sum of squared variables $X_k^* X_k$. By Parseval's theorem, the sum of the spectral density is equal to the total variance, which we've already seen is a χ^2 variable,

$$\sum_k A_k^2 \Delta f = \hat{\sigma}^2 \sim \frac{\sigma^2}{\nu} \chi_\nu^2, \quad \text{with } \nu = N.$$

So it stands to reason, that each $S(f_k) = A_k^2$ is a χ^2 statistic, that is,

$$A_k^2 \sim \frac{\sigma_k^2}{\nu} \chi_\nu^2, \quad \text{with } \nu = 2. \quad (30)$$

There are two degrees of freedom in the estimate of A_k because we added the positive and negative frequencies, recall that for a real input record,

$$A_{k>0}^2 = 2X_k^* X_k = (X_k^* X_k + X_{-k}^* X_{-k}).$$

With the above considerations, the appropriate probability statement becomes,

$$\text{Prob}\left[\frac{\chi_{\nu, 1-\alpha/2}^2}{\nu} \leq \frac{\hat{S}(f)}{S(f)} \leq \frac{\chi_{\nu, \alpha/2}^2}{\nu}\right] = 1 - \alpha, \quad (31)$$

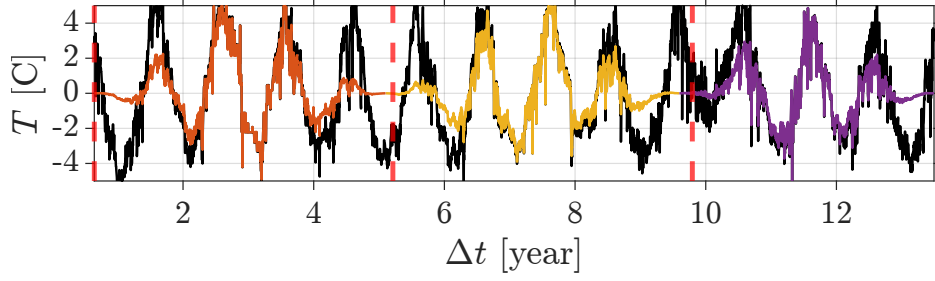


Figure 14: Detrended SIO-pier temperature record (black) with 4-year non-overlapping sub-windows (red dashed) and scaled by the Hann-function (colors).

with the 100%(1 − α) confidence interval given by,

$$\frac{\nu}{\chi_{\nu,\alpha/2}^2} \hat{S}(f) \leq S(f) \leq \frac{\nu}{\chi_{\nu,1-\alpha/2}^2} \hat{S}(f). \quad (32)$$

The CI has a constant height on a log-scale plot, because the upper- and lower-bounds become,

$$\begin{aligned} \log \left(\frac{\nu}{\chi_{\nu,1-\alpha/2}^2} \hat{S}(f) \right) &= \log \left(\frac{\nu}{\chi_{\nu,1-\alpha/2}^2} \right) + \log \left(\hat{S}(f) \right), \\ \log \left(\frac{\nu}{\chi_{\nu,\alpha/2}^2} \hat{S}(f) \right) &= \log \left(\frac{\nu}{\chi_{\nu,\alpha/2}^2} \right) + \log \left(\hat{S}(f) \right), \end{aligned}$$

respectively. So, instead of plotting the CI as bands above/below the spectral amplitude $S(f)$, we need to only include it at a single frequency (black, Figure 13). Spectral peaks/valleys that have a height that exceeds that of the CI are considered statistically significant. The 2-dof spectra are very fuzzy. We can improve this by averaging over sub-windows.

4.4 Welch's Method

We can increase the number of degrees of freedom of our spectral estimate by splitting the original record into M -subsets, computing spectra on each subset, and averaging the M -estimates of each amplitude A_k^2 . The first step is to determine the number of sub-windows M . There is a trade-off in spectral resolution because each sub-window m will have only $N_s = \text{floor}(N/M)$ samples (no overlap in subwindows). So, it is vital to first estimate the relevant time-scales within your signal, and choose a subwindow size that's several times this duration. For example, if you're confident that the dominant variability is annual (black, Figure 14), a subwindow of length $\mathcal{T} = 4$ year should resolve the peak at one-cycle-per-year. For the m^{th} sub-window, the spectral amplitudes are,

$$\hat{S}_m(k) = \hat{A}_m^2(k) = 2\hat{X}_m^*(k)\hat{X}_m(k), \quad \text{for } m = \{1, 2, \dots, M\}, \quad (33)$$

where,

$$\hat{X}_m(k) = \sum_{n=1+(m-1)N_s}^{mN_s} x_n \exp \left(-i \frac{2\pi k}{N_s} n \right), \quad (34)$$

the $\hat{}$ indicates that these are estimates and the factor of 2 is for real input-signal $x(t) = x^*(t)$. The average across the subwindows gives our M -point estimate of the spectral amplitude,

$$\hat{S}(k) = \frac{1}{M} \sum_{m=1}^M \hat{S}_m(k). \quad (35)$$

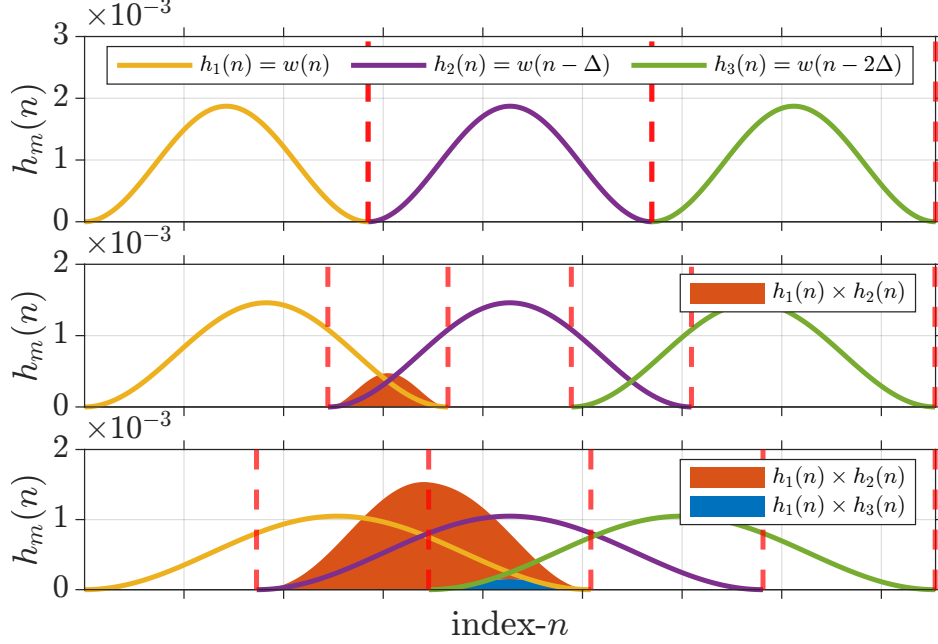


Figure 15: Examples of non-overlapping window (top), $\beta = 1/3$ overlap (middle), and $\beta = 2/3$ overlap for a Hann taper function (colored lines). Also shown as colored areas are scaled products of the overlapping windows $h_1(n) \times h_2(n)$ (orange) and $h_1(n) \times h_3(n)$ blue. The covariance increases with increasing overlap.

This is commonly called Welch's Method. By averaging the spectral estimates of each subwindow, we increase the number of degrees of freedom to $\nu = 2M$.

As an added level of sophistication, we can partially overlap the sub-windows. That is, instead of offsetting each sub-window by N_s -indices, we can shift by $\Delta = (1 - \beta)N_s$ so that the windows overlap by $\beta \times 100\%$. For a chosen number of sub-windows M , overlapping also increases the number of elements to,

$$N_s = \text{floor} \left(\frac{N - 1}{M(1 - \beta) + \beta} \right) > \frac{N}{M},$$

in each sub-window, which improves spectral resolution. The Fourier transform of the m^{th} subwindow becomes,

$$\hat{X}_m(k) = \sum_{n=1+(m-1)\Delta}^{(m-1)\Delta+N_s} x_n \exp \left(-i \frac{2\pi k}{N_s} n \right), \quad (36)$$

This is particularly advantageous because, the taper function damps information at the start/end of each window (Figure 14). So, the overlap ensures this information is used in the spectral estimate (*e.g.*, colored lines, Figure 15).

The number of degrees of freedom also increase $\nu > 2N/N_s$. However, there is redundant information (covariance) in the overlapping portions of each sub-window (colored areas, Figure 15), making the derivation of ν rather daunting. That said, the derivation is rooted in the now familiar method of determining the expected value and variance of the estimator $\hat{S}$. The resulting expression for the number of degrees of freedom is,

$$\begin{aligned} \nu &= 2 \frac{E[\hat{S}]^2}{V[\hat{S}]}, \\ &= \frac{2M}{1 + 2 \sum_{m=1}^{M-1} \left(1 - \frac{m}{M}\right) |\rho(m)|^2}, \end{aligned}$$

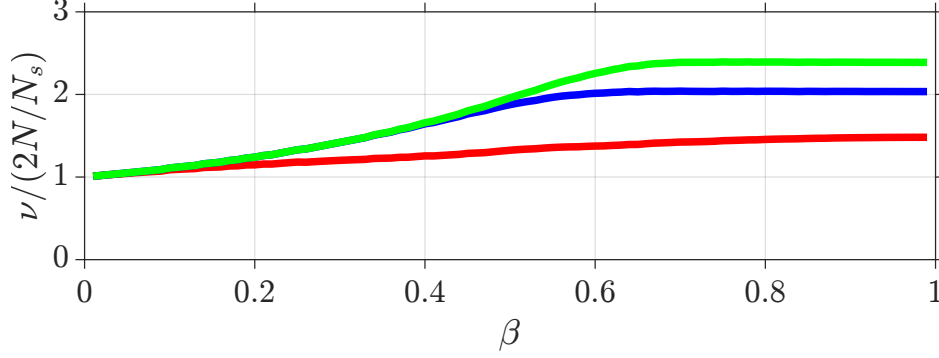


Figure 16: Estimated degrees of freedom ν versus fractional window overlap β for a box-car window (red), Hann taper function (blue) and Blackman taper function (green).

where the window auto-correlation function is,

$$\rho(m) = \sum_{n=1}^{N_s-m\Delta} h(n)h(n+m\Delta), \quad \text{with } h(n) = \frac{w(n)}{(\sum_n |w(n)|^2)^{1/2}},$$

such that at zero-lag $\rho(0) = 1$ and outside the sub-window $\rho(m) = 0$ because there is no overlap wherever $m\Delta \geq N_s$. The overlapping degrees of freedom increase monotonically with increasing fractional overlap β relative to $2N/N_s$ (Figure 16). The maximum degrees of freedom asymptote for cosine taper windows around $\beta \geq 2/3$ at $\nu \approx 2(2N/N_s)$ and $\nu \approx 2.3(2N/N_s)$ for Hann and Blackman taper functions, respectively (blue and green curves, Figure 16). It pays to overlap!

Here is code for computing the spectra in Figure 13:

```
%% LaTeX2e file 'SIOpier_spectra.m'
%% generated by the 'filecontents' environment
%% from source 'Lecture07' on 2023/11/22.
%%
% input variables from SIO-pier historical record:
% t = elapsed time in decimal years
% dt = sample interval in decimal years
% Ts = detrended surface temperature record (daily observations)
% N = number of observations in record
%
M = 16;
lap = 2/3;
[A2 , f ] = welch_method(Ts,dt,N,1,0);% function at end-of-file
[A2w,fw] = welch_method(Ts,dt,N,M,lap);
%
% previously , we estimated that \nu/(2N/Ns)=2
% for overlap=2/3 with Hann-function taper
dof = 2;
Xp = chi2inv(0.9,dof);
Xm = chi2inv(0.1,dof);
err = (A2w(end)*[dof/Xp dof/Xm]);
%
dof = 4*M;
Xp = chi2inv(0.9,dof);
Xm = chi2inv(0.1,dof);
err_w = (A2w(end)*[dof/Xp dof/Xm]);
%
% generate a figure and axes
```

```

xm = 2.5; ym = 2; pw = 10; ph = 6;
ppos1=[xm ym pw ph];
cbpos=[xm+0.8*pw ym+0.25*ph ag/2 ph/5];
ps = [1.5*xm+pw 1.5*ym+ph];
%
% create figure
fig0= figure('units','centimeters','color','w');
pos = get(fig0,'position');
pos(3:4)=ps;
set(fig0,'position',pos,'papersize',ps,'paperposition',[0 0 ps])
f0ax1 = axes('units','centimeters','position',ppos1);
f0ax1p1=loglog(f0ax1,f,A2,'-k',fw,A2w,'-r','linewidth',2);hold on
f0ax1p2=loglog(f0ax1,f(2)*[1 1],err,'s-k',2*f(2)*[1 1],err_w,'s-r','
    linewidth',2,'markersize',3)
ylabel(f0ax1,'$S_{\scriptscriptstyle T}$ [C$^2$/Hz]','interpreter','latex')
xlabel(f0ax1,'$f$ [cycle/year]','interpreter','latex')
grid on
set(f0ax1,'tickdir','out','ticklabelinterpreter','latex','ylim',[1e-7 1e2],...
    'xlim',[0.5*f(2) f(end)],'xtick',[1e-2 1e-1 1e0 1e1 1e2])
%
function [A2,f] = welch_method(x,dt,N,M,olap);
% x = input record of length N
% dt = sample interval
% M = number of sub-windows
% olap = fractional overlap of sub-windows
%
Ns = floor(N/(M*(1-olap)+olap));% Ns elements in window
ds = floor(Ns-olap*Ns);% shift for start of sub-window
%
% reduced frequency resolution
f0 = 1/(Ns*dt);
%
% Positive frequencies are different for even vs. odd N
if iseven(Ns)
    stop = Ns/2;
else
    stop = (Ns+1)/2;
end
%
f = f0*[0:stop-1];
%
% each window will be tapered
win = hanning(Ns);
% pre-allocate the fft array
F = nan(stop,M);
for m = 1:M
    n = [1:Ns] + (m-1)*ds;% indices in subwindow
    xp = x(n);% data in sub-window
    s2 = var(xp);% untapered window variance
    xw = detrend(xp).*win;% detrend+apply taper
    xw = xw*s2/var(xw);% recover variance
    Xw = fft(xw);% fast Fourier transform
    F(:,m) = Xw(1:stop);% log positive frequencies
end

```

```

% get real/imaginary parts
C = real(F);
Q = imag(F);
%
% average the spectra esimated from each window
A2 = 2*sum( (C.^2 + Q.^2) ,2)*(dt/(Ns*M));% units=[C^2/Hz]
end

```